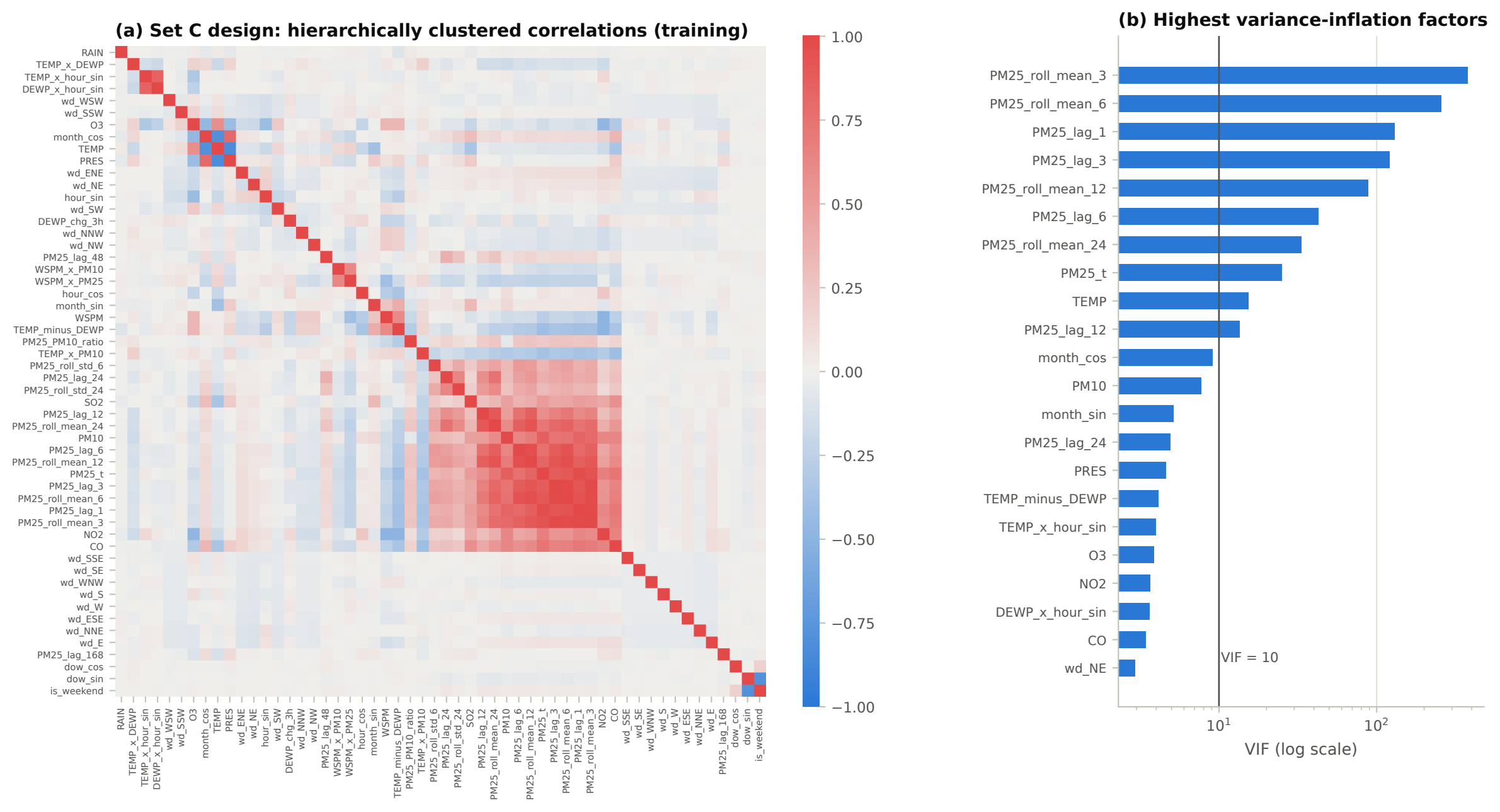
(a) Set C design: hierarchically clustered correlations (training)
RAIN
TEMP_x_DEWP
TEMP_x_hour_sin
DEWP_x_hour_sin
wd_WSW
wd_SSW
O3
month_cos
TEMP
PRES
wd_ENE
wd_NE
hour_sin
wd_SW
DEWP_chg_3h
wd_NNW
wd_NW
PM25_lag_48
WSPM_x_PM10
WSPM_x_PM25
hour_cos
month_sin
WSPM
TEMP_minus_DEWP
PM25_PM10_ratio
TEMP_x_PM10
PM25_roll_std_6
PM25_lag_24
PM25_roll_std_24
SO2
PM25_lag_12
PM25_roll_mean_24
PM10
PM25_lag_6
PM25_roll_mean_12
PM25_t
PM25_lag_3
PM25_roll_mean_6
PM25_lag_1
PM25_roll_mean_3
NO2
CO
wd_SSE
wd_SE
wd_WNW
wd_S
wd_W
wd_ESE
wd_NNE
wd_E
PM25_lag_168
dow_cos
dow_sin
is_weekend
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
(b) Highest variance-inflation factors
PM25_roll_mean_3
PM25_roll_mean_6
PM25_lag_1
PM25_lag_3
PM25_roll_mean_12
PM25_lag_6
PM25_roll_mean_24
PM25_t
TEMP
PM25_lag_12
month_cos
PM10
month_sin
PM25_lag_24
PRES
TEMP_minus_DEWP
TEMP_x_hour_sin
O3
NO2
DEWP_x_hour_sin
CO
wd_NE
VIF = 10
$10^1$
$10^2$
VIF (log scale)